

PHINMA UNIVERSITY OF ILOILO

College of Information Technology Education

CAPSTONE PROJECT PROPOSAL

Check-In and Billing with POS Integration: A Smart Operations System for IdeaHub Co-working Space + Café

In partial fulfillment for the requirements in Capstone 1 and Research Subject

RAMOS, RYBELLE V. BSIT3-6

OÑANG, CARL NATHANIEL BSIT3-6

PROTACIO, JERALYN M. BSIT3-6

DELOS REYES, AJ KURT BSIT3-6

ROCKWELL, SAVER BSIT3-6

System's Development Track

Second Semester

2025-2026

CHAPTER 1

INTRODUCTION

1.1 Background of the Study

The continuous advancement of information technology has significantly transformed the operational processes of service-oriented businesses such as cafés, co-working spaces, and shared commercial establishments. Businesses are increasingly adopting automated systems to improve efficiency, reduce operational errors, enhance customer experience, and support data-driven decision-making. Modern digital solutions such as Point of Sale (POS) systems, inventory management systems, analytics dashboards, and automated billing platforms have become essential tools for improving business productivity and operational control.

Despite these technological advancements, many small and medium-sized enterprises still rely on manual or semi-computerized methods for handling daily business operations. In establishments like IdeaHub Co-working Space + Café, processes such as customer check-in and check-out, billing computation, inventory monitoring, sales tracking, employee attendance, and operational reporting are often managed manually. These traditional processes consume time, increase the risk of human error, create inconsistencies in records, and limit the management's ability to monitor business performance in real time.

Manual billing and session monitoring may lead to inaccurate customer charges, delayed transactions, and inefficient customer service. Likewise, inventory monitoring without automated notifications can result in unnoticed stock shortages or inconsistencies in supply management. The absence of a centralized reporting and analytics system also makes it difficult for business owners to evaluate staff productivity, monitor financial performance, and generate accurate daily operational reports.

To address these challenges, this capstone project proposes the development of an Integrated Smart Operations Management System with POS Integration for IdeaHub Co-working Space + Café. The system is designed to automate customer check-in and billing processes, integrate café order management, provide inventory monitoring with in-system notifications, generate downloadable reports, support loan service and product management, implement daily AM-to-PM soft balancing, and provide staff performance analytics through a centralized web-based platform.

The proposed system aims to improve operational efficiency, billing accuracy, inventory management, financial monitoring, and employee productivity tracking while providing management with real-time business insights. By utilizing modern web technologies, database management systems, and real-time processing mechanisms, the study seeks to provide a reliable and scalable solution that enhances both business operations and customer service quality.

1.2 Statement of the Problem

The current operational processes of IdeaHub Co-working Space + Café rely heavily on manual and semi-computerized procedures. These processes result in inefficiencies, inaccurate records, delayed monitoring, and difficulties in managing daily business operations. The lack of a centralized management system affects transaction processing, inventory control, reporting, and staff performance monitoring.

Specifically, the study seeks to answer the following problems:

1. How can the system automate customer check-in and check-out processes to improve operational efficiency?
2. How can the system reduce billing inaccuracies and improve transaction computation through POS integration?
3. How can the system improve inventory monitoring and stock management through automated inventory tracking?
4. How can the system provide in-system inventory notifications for low-stock items and inventory concerns?
5. How can the system implement daily AM-to-PM soft balancing for accurate operational and financial monitoring?
6. How can the system generate organized and downloadable reports for sales, inventory, transactions, and operational records?
7. How can the system manage loaned products and services to improve accountability and monitoring?
8. How can the system monitor and analyze staff productivity and performance through analytics features?
9. How can the system provide real-time operational monitoring and centralized data management?
10. How can the system reduce manual errors and improve the accessibility of operational information for business decision-making?

1.3 Objectives of the Study

1.3.1 General Objectives

The general objective of the study is to design and develop an Integrated POS, Inventory, Analytics, and Operational Management System for IdeaHub Co-working Space + Café that automates customer services, billing operations, inventory monitoring, operational balancing, reporting, and staff performance tracking to improve business efficiency and operational management.

1.3.2 Specific Objectives

Specifically, the study aims to:

1. Develop a system that automates customer check-in and check-out processes.
2. Implement an integrated Point of Sale (POS) and billing module for accurate transaction processing.
3. Design an inventory management module capable of monitoring stock levels and inventory movement.
4. Develop an in-system inventory notification feature for low-stock monitoring.
5. Implement a daily AM-to-PM soft balancing feature for monitoring operational transactions and cash flow.
6. Create downloadable reports for sales, transactions, inventory records, and operational summaries.
7. Develop a loan service and product management feature for tracking borrowed items and monitoring their return status.
8. Design a staff analytics module for monitoring employee productivity and operational performance.
9. Provide a centralized real-time dashboard for monitoring transactions, inventory status, customer sessions, and operational activities.
10. Improve operational efficiency, reduce manual errors, and support better business decision-making through automation.

1.4 Scope and Delimitation

This study focuses on the development of a web-based Integrated Smart Operations Management System with POS Integration for IdeaHub Co-working Space + Café. The system is designed to automate and centralize business operations including customer check-in and check-out, time-based billing, POS transactions, café order management, inventory monitoring, loan service tracking, staff performance monitoring, and operational reporting.

The system allows authorized users such as administrators and staff members to manage customer sessions, process orders and billing, monitor inventory levels, receive stock notifications, generate downloadable reports, perform daily AM-to-PM soft balancing, and track employee performance through analytics dashboards.

The system includes the following major modules:

- Customer Session Management
- POS and Billing Management
- Inventory Monitoring with Notifications
- Loan Services/Amount Receivables
- Staff Attendance and Performance Analytics
- Daily Soft Balancing
- Real-Time Dashboard and Reporting
- User and Access Management

The study is limited to internal operational use within IdeaHub Co-working Space + Café only. The system will operate as a web-based platform and will not include mobile application support, online payment gateways, artificial intelligence prediction systems, or integration with third-party accounting platforms. Advanced financial forecasting and external cloud-based synchronization are also outside the scope of the study.

1.5 Significance of the Study

The proposed system will benefit the following:

- **Business Owners (IdeaHub Management)** – The proposed system will help business owners monitor operations more efficiently through centralized dashboards, automated reporting, inventory monitoring, and operational analytics. The system will provide real-time access to sales records, transaction summaries, staff performance reports, and inventory data, allowing owners to make informed decisions and improve operational management.
- **Staff/Employees** – The system will assist staff members in handling customer transactions, monitoring inventory, processing billing operations, and managing daily operational activities more efficiently. Automated features will reduce manual workload, minimize errors, and improve productivity.
- **Customers** – Customers will benefit from faster transaction processing, more accurate billing, improved service quality, and organized customer session management. The system is expected to improve the overall customer experience within the establishment.
- **BSIT Students** – The project demonstrates the practical application of web development, database management, operational automation, analytics systems, and integrated information systems within a real-world business environment.
- **Future Researchers** – The study may serve as a reference for future researchers who plan to develop similar business management systems, POS systems, inventory systems, analytics platforms, or operational monitoring systems.

1.6 Definition of Terms

- **Application Development** – The process of designing, building, and deploying software applications.
- **Point of Sale (POS) System** – A computerized system used to process transactions, record sales, and manage business operations.
- **Billing System** – A component of the system that calculates and generates the total amount to be paid based on services and products used.
- **Check-In/Check-Out System** – A feature that records the time a customer enters and exits the establishment for accurate session tracking.
- **Time-Based Billing** – A method of charging customers based on the duration of their stay or usage of services.
- **Session Management** – The process of tracking customer activity, including time-in, time-out, and usage duration.
- **Order Management** – The system function that handles customer orders, including food and beverage transactions.
- **Dashboard** – A user interface that displays real-time information such as active sessions, sales, and system status.
- **Real-Time Monitoring** – The ability of the system to update and display data instantly as transactions occur.
- **Database Management** – The process of storing, organizing, and retrieving data efficiently using a database system.
- **User Authentication** – A security process that verifies the identity of users before granting access to the system.
- **Administrator** – A user responsible for managing the system, including monitoring operations and maintaining data.
- **Web-Based System** – A system that operates through a web browser and is accessible over a network.
- **Automation** – The use of technology to perform tasks with minimal human intervention, reducing errors and improving efficiency.

CHAPTER 2

REVIEW OF RELATED LITERATURE AND STUDIES

2.1 Foreign Studies

Secure and Efficient POS (Point of Sale) Billing Software

A study by Azhagusundaram et al. (2025) focused on the development of a secure and efficient POS billing system designed to improve transaction processing in retail and service industries. The research highlighted that traditional billing systems often suffer from slow processing, security vulnerabilities, and integration issues. The proposed system uses modern technologies such as secure authentication, encryption, and cloud-based architecture to provide fast, secure, and reliable billing operations. The study concluded that advanced billing systems significantly improve efficiency, security, and customer satisfaction.

2.2 Local Studies

Coffee Shop Owners' Decision and Use of Point-of-Sale (POS) System

Several local studies have examined the adoption of point-of-sale (POS) systems among coffee shop businesses in the Philippines. A study conducted in Baliwag City analyzed the perceptions of coffee shop owners regarding the use of POS systems and found that these systems improve transaction processing, inventory management, and customer service. However, the study also revealed that factors such as cost, system complexity, and employee readiness affect the adoption of such technologies. Despite these challenges, most business owners recognize the importance of POS systems in enhancing efficiency and maintaining competitiveness. These findings highlight the need for accessible and user-friendly POS solutions to support small businesses in improving their operations and adapting to modern technological advancements.

2.3 Related Literature

Application of Billing Pos (point of Sales) on Cloud Computing for Restaurant by Using IoT (internet of Things) Device with Data Synchronized Method in The Reporting Process

The study by Adityo et al. (2018), entitled "Application of Billing POS (Point of Sales) on Cloud Computing for Restaurant by Using IoT (Internet of Things) Device with Data Synchronized Method in The Reporting Process," designed and developed a POS billing system for restaurant taxpayers using IoT devices, specifically the Raspberry Pi 3, integrated with Cloud Computing technology via Google Cloud Platform. The system aimed to address problems such as tax leakage and manual bookkeeping by enabling real-time transmission of sales data to a government revenue server. Results showed that the system achieved an average data transmission speed of 0.7833 seconds per transaction, proving it to be effective and efficient in monitoring revenue in real time.

2.4 Synthesis of Reviewed Literature

The reviewed foreign and local studies, along with related literature, highlight the significant role of Point of Sale (POS) and billing systems in improving business operations and efficiency. The study by Azhagusundaram et al. (2025) emphasizes that modern POS billing systems enhance transaction speed, security, and reliability through advanced technologies. Similarly, Boado et al. (2025) found that POS systems improve transaction processing, inventory management, and customer service in local coffee shop businesses, although challenges in adoption still exist. In addition, Adityo et al. (2018) demonstrated that integrating POS billing systems with cloud computing and IoT enables real-time monitoring and accurate reporting. These findings indicate the need for an integrated and user-friendly POS and billing system that can support both time-based and product-based transactions, which the proposed system aims to address.

3.0 References

Azhagusundaram, R., Soundhar, K., Sivakumar, D., Sasi Kumar, M., & Saravanakumar, J. (2025). Secure and efficient POS (Point of Sale) billing software. *International Journal of Innovative Research in Technology (IJIRT)*, 11(11).

https://ijirt.org/publishedpaper/IJIRT176066_PAPER.pdf

Boado, R. J. R., Francisco, J. M. D., Lingat, V. N. R., Reobaldo, R. A. D., et al. (2025). Coffee shop owners' decision and use of Point-of-Sale (POS) system. *International Journal of Research Publication and Reviews*, 6(3), 2092–2102.

<https://doi.org/10.55248/gengpi.6.0325.1170>

Adityo, R & Miawarni, Herti. (2018). Application of Billing Pos (point of Sales) on Cloud Computing for Restaurant by Using IoT (internet of Things) Device with Data Synchronized Method in The Reporting Process. *JEECS (Journal of Electrical Engineering and Computer Sciences)*. 3. 431-438. 10.54732/jeecs.v3i2.130.

CHAPTER 3

METHODOLOGY

3.1 Research Design

This study uses a **developmental research design** to develop a Check-in and Billing System with Point of Sale (POS) Integration for IdeaHub Co-working Space + Café. This design focuses on the creation, design, and evaluation of a web-based system intended to improve business operations, including customer session management, time-based billing, order processing, and sales monitoring. The system will be developed based on the identified problems in manual operations and the specific requirements of the business to achieve accuracy, efficiency, and usability.

3.2 System Development Methodology

This study will use the **Agile methodology** in developing the Check-in and Billing System with Point of Sale (POS) Integration for IdeaHub Co-working Space + Café, consisting the following phases:

1. Requirements Analysis
2. System Design
3. Development
4. Testing
5. Deployment (Prototype)
6. Continuous Improvement

3.3 System Architecture

The Integrated Smart Operations Management System with POS Integration follows a client-server architecture combined with a three-tier system structure consisting of the presentation layer, application layer, and data layer.

The presentation layer serves as the user interface where administrators and staff interact with the system using a web browser. This layer is responsible for displaying dashboards, reports, transaction forms, inventory data, customer sessions, and analytics information while capturing user inputs.

The application layer contains the core business logic of the system. Developed using Flask Framework and Python technologies, this layer processes customer check-ins and check-outs, computes billing, handles POS transactions, monitors inventory levels, manages loan services, performs daily soft balancing, generates downloadable reports, and calculates staff performance analytics. It also manages authentication, authorization, and real-time operational monitoring.

The data layer utilizes MySQL Database Management System with SQLAlchemy ORM for structured data storage, retrieval, and management. This layer stores customer records, transactions, inventory data, employee attendance, sales reports, loan transactions, and analytics information.

The architecture enables centralized data management, secure multi-user access, real-time monitoring, efficient transaction processing, scalability, and improved business decision-making within the operational environment of IdeaHub Co-working Space + Café.

3.4 Tool and Technology Used

The development of the Integrated POS, Inventory, Analytics, and Operational Management System for IdeaHub Co-working Space + Café utilized various software technologies, programming languages, frameworks, and development tools to achieve efficient system performance, real-time processing, database management, and user accessibility.

Frontend Technologies

- **HTML (HyperText Markup Language)** - HTML was used as the primary structure of the web-based system. It was utilized in creating the layout, forms, dashboards, tables, navigation menus, and other interface components of the system. HTML provides the structural foundation of the user interface and helps with proper organization of web elements.
- **CSS (Cascading Style Sheets)** - CSS was used to enhance the visual appearance and responsiveness of the system interface. It was utilized for designing dashboards, navigation panels, buttons, tables, forms, and reports to improve user experience and accessibility. CSS also helps in consistency in the overall design and layout of the system.
- **JavaScript (JS)** - JavaScript was used to implement dynamic functionalities and interactive features within the system. It was utilized for real-time updates, input validations, automated computations, notifications, interactive dashboards, and operational responsiveness. JavaScript improved the efficiency and usability of the system by allowing faster interaction between users and the interface.

Backend Technologies

- **Python** - Python served as the primary backend programming language used in developing the system functionalities and business logic. Python was selected because of its flexibility, readability, reliability, and ability to handle operational processing efficiently.
- **Flask Framework** - Flask is a lightweight Python web framework used for developing the server-side functionalities of the system. Flask was utilized to manage routing, authentication, transaction processing, operational logic, report generation, and communication between the frontend and database. The framework also supports scalability and modular system development.

Database Technologies

- **MySQL** - MySQL was used as the relational database management system for storing and managing operational data. The database stores customer records, transactions, inventory data, loan records, balancing reports, staff analytics, and system logs. MySQL was selected because of its reliability, performance, and capability to manage structured data efficiently.

- **SQLAlchemy** - was used as the Object Relational Mapper (ORM) for database interaction. It simplifies database operations by allowing developers to interact with database tables using Python objects and queries instead of manually writing complex SQL statements. SQLAlchemy improves database management, maintainability, and system scalability.

Development Tools

- **Visual Studio Code (VS Code)** - was used as the primary integrated development environment (IDE) during system development. It provided tools for coding, debugging, extension integration, syntax highlighting, and project management. VS Code improved development efficiency and supported collaboration among proponents.
- **GitHub** - was used for version control and project repository management. It allowed the proponents to organize source code, monitor revisions, and maintain backup copies of the project throughout the development process.
- **XAMPP** - was utilized as the local server environment during system development and testing. It provided local hosting support and database management capabilities necessary for running the web-based system.

3.5 Data Gathering Techniques

The proponents utilized several data gathering techniques to identify the operational problems, business requirements, and necessary functionalities needed for the development of the proposed system. These techniques helped the proponents understand the current workflow and operational challenges experienced by IdeaHub Co-working Space + Café.

Interview with Clients and Business Owners

The proponents conducted interviews with the business owners and staff members of IdeaHub Co-working Space + Café to gather information regarding the current operational procedures, challenges, and system requirements. The interviews focused on customer check-in processes, billing operations, inventory management, staff monitoring, balancing procedures, and reporting concerns. The interview also allowed the proponents to identify the features and functionalities needed to improve operational efficiency and support business management.

Observation of Existing Operational Processes

The proponents observed the current operational setup of the establishment, including customer check-in and check-out procedures, billing computations, POS transactions, inventory monitoring, and operational balancing activities. Observation helped the proponents identify inefficiencies, delays, manual errors, and operational difficulties encountered during daily business operations. Through observation, the proponents were able to analyze the workflow and determine how automation could improve operational performance.

Review of Existing Check-In and Billing with POS System

The proponents reviewed the existing check-in and billing procedures currently implemented within the establishment. The review focused on how customer sessions were monitored, how transactions were processed, and how billing computations were performed. This review helped identify limitations in the current process such as inaccurate billing, delayed monitoring, inconsistent transaction records, and lack of centralized operational management.

Collection and Review of Existing Menu and Operational Data

The proponents requested copies of the existing menu, pricing information, inventory records, and operational data from the establishment. These records were used as references during system design and database development. The collected information helped in the proposed system accurately reflects the operational requirements and business processes of the establishment.

Documentation Review

The proponents reviewed existing operational documents, transaction records, inventory logs, and reporting procedures to further understand the current management process of the business. The review also served as a basis for designing the system modules and operational workflows.

3.6 Testing Procedures

Unit Testing

Unit testing was conducted to evaluate the functionality of individual system modules and components. Each module was tested independently to achieve proper operation and accurate processing.

The following modules underwent unit testing:

- User Authentication Module
- POS and Billing Module
- Inventory Management Module
- Inventory Notification Module
- Loan Service Management Module
- Daily Soft Balancing Module
- Staff Analytics Module
- Report Generation Module

Unit testing guarantees that each module functioned correctly before integration into the complete system.

System Testing

System testing was conducted to evaluate the overall performance and integration of all system modules. This testing procedure guarantees that the system operates as a complete and unified platform.

The testing focused on:

- Transaction processing
- Inventory monitoring
- Billing computation
- Report generation
- Dashboard monitoring
- User authentication
- Operational balancing
- Real-time updates

System testing also identified compatibility issues, operational errors, and performance concerns that required improvement.

User Acceptance Testing (UAT)

User Acceptance Testing was conducted with selected users including business owners, staff members, and system users to evaluate the usability, effectiveness, and functionality of the system in an actual operational environment.

The respondents tested the system functionalities and provided feedback regarding:

- Ease of use
- Accuracy of billing
- Efficiency of transactions
- Accessibility of reports
- Inventory monitoring
- Dashboard usability
- Operational performance

The feedback gathered during User Acceptance Testing was used to improve system functionality and user experience before final deployment.

Performance Testing

Performance testing was conducted to evaluate the responsiveness, processing speed, and operational stability of the system during multiple transactions and simultaneous operations.

This testing guarantees that the system could efficiently handle operational activities without significant delays or performance issues.

Security Testing

Security testing was conducted to verify the effectiveness of authentication, access control, session management, and data protection mechanisms implemented within the system.

This testing helped assuring that operational records and user information remain protected from unauthorized access.

3.7 Evaluation Criteria

The proposed system will be evaluated based on the following criteria to determine its effectiveness, usability, reliability, and operational performance.

Functionality

The system will be evaluated based on its ability to perform the required operational tasks and functionalities accurately and efficiently. This includes transaction processing, inventory monitoring, report generation, balancing operations, loan management, and analytics monitoring.

Usability

The system will be evaluated based on its ease of use, user-friendliness, interface accessibility, and overall user experience. The evaluation will determine whether users can navigate and operate the system efficiently with minimal difficulty.

Accuracy of Billing and Transaction Computation

The system will be evaluated based on the accuracy of billing computations, including customer session charges, food orders, operational fees, and total transaction calculations.

The evaluation will determine whether the system correctly computes:

- Time-based billing
- Food and beverage charges
- Total customer bills
- Daily balancing computations
- Sales summaries

Security

The system will be evaluated based on the effectiveness of its security measures including user authentication, password protection, access control, session management, and data protection.

Efficiency

The system will be evaluated based on its ability to improve operational workflows, reduce processing time, minimize manual workload, and improve transaction speed and operational monitoring.

Reliability

The system will be evaluated based on its consistency, operational stability, and ability to perform continuously without significant errors or failures.

Real-Time Monitoring Capability

The system will be evaluated based on its capability to provide real-time updates and monitoring of transactions, inventory levels, customer sessions, and operational activities.

Inventory Monitoring Accuracy

The system will be evaluated based on the accuracy of inventory tracking, stock updates, and inventory notification features.

Staff Analytics Effectiveness

The system will be evaluated based on its ability to monitor employee productivity, operational activities, and staff performance accurately.

User Satisfaction

The system will be evaluated based on the feedback and satisfaction level of users regarding the overall performance, usability, and effectiveness of the system in supporting business operations.

SUPPORTING DIAGRAMS

DATA FLOW DIAGRAM

User → Check-In and Billing System → Database → User

- Users (staff/admin) input customer details, orders, and bookings through the system
- The system processes customer sessions, billing computations, order management, and boardroom bookings
- Processed data is stored, updated, and retrieved from the database
- The system returns real-time information such as active sessions, total bills, transactions, and booking schedules.

ENTITY RELATIONSHIP DIAGRAM

Entities:

- User (UserID, Username, Password, Role, CreatedAt)
- CustomerSession (SessionID, CustomerName, School, Course, SpaceTypeID, TimeIn, TimeOut, Status)
- SpaceType (SpaceTypeID, Name, RatePerMinute)
- Order (OrderID, SessionID, CreatedAt)
- OrderItem (OrderItemID, OrderID, MenuItemID, Quantity, Price)
- MenuItem (MenuItemID, Name, Price, Category, Status)
- Transaction (TransactionID, SessionID, TimeBill, FoodBill, TotalBill, CreatedAt)
- BoardroomBooking (BookingID, CustomerName, Date, StartTime, EndTime, NumberOfPeople, Purpose, Status)
- StaffAttendance (AttendanceID, UserID, TimeIn, TimeOut)

Relationships:

- A **User** manages system operations
- A **CustomerSession** belongs to a **SpaceType**
- A **CustomerSession** can have multiple **Orders** (one-to-many)
- An **Order** contains multiple **OrderItems** (one-to-many)
- Each **OrderItem** is linked to a **MenuItem** (one-to-one)
- A **CustomerSession** generates a **Transaction** after checkout (one-to-one)
- A **BoardroomBooking** represents scheduled usage of the boardroom
- A **User** can have multiple **StaffAttendance** records (one-to-many)

USE CASE DIAGRAM

Actors:

- Staff / User
- Administrator

Use Cases:

Staff / User:

- Log In to the System
- Check-in Customer
- View Active Sessions
- Add Orders to Customer
- View and Manage Orders
- Void Incorrect Items
- Process Checkout and Billing
- Book Boardroom
- View Boardroom Bookings

Administrator:

- Log In to the System
- Manage Menu Items
- View Sales Reports
- Monitor Transactions
- Manage User Accounts
- Oversee System Operations
- Log In to the System